

Toy Store Sales Analytics Dashboard

1. Project Overview

This project is an end-to-end Business Intelligence solution built using SQL and Power BI to analyze an e-commerce toy store dataset. The objective was to transform raw transactional data into actionable business insights through interactive dashboards.

Business Scenario: Management required a centralized dashboard to monitor revenue, orders, product performance, website traffic, marketing effectiveness, conversion rate and refund trends.

2. Business Problem Statement

Business data was distributed across multiple tables, making analysis slow and manual. A unified reporting solution was required.

3. Business Objectives

- Build interactive Power BI dashboards
- Monitor KPIs
- Analyze sales, products and marketing
- Support business decisions

4. Dataset Overview

Orders, Order Items, Products, Website Sessions, Website Pageviews, Order Item Refunds.

5. Data Relationships

Orders↔Order Items, Orders↔Sessions, Products↔Order Items, Orders↔Refunds.

6. SQL Analysis Summary

Performed data exploration, sales, product, website and marketing analysis.

7. Power BI Development

Imported CSVs, Power Query cleaning, data model, DAX measures, dashboard creation.

8. Data Model

Orders acts as the central fact table.

9. DAX Measures

Total Revenue, Orders, Sessions, Conversion Rate, Average Order Value, Refund Amount.

10. Executive Dashboard

KPIs: Revenue \$2M, Orders 32K, Sessions 473K, Conversion Rate 6.83%, AOV \$60, Refund Amount \$85K, Refunds 2K.

11. Product Performance Dashboard

Analyzes revenue, units sold, average order value and revenue contribution by product.

12. Marketing & Website Analytics Dashboard

Analyzes sessions, orders, revenue by traffic source and device type.

13. Business Metrics & Key Findings

- Generated approximately \$2M revenue from 32K orders.
- 473K sessions with a 6.83% conversion rate.
- Average Order Value was \$60.
- Refund amount totaled \$85K (about 4.25% of revenue).
- Mr. Fuzzy generated \$1.21M revenue, sold 24,226 units and contributed 62.47% of revenue.
- Google Search generated 316K sessions, 21,333 orders and ~\$1.3M revenue.
- Desktop users accounted for 69.16% of website sessions.

14. Business Recommendations

- Increase investment in Google Search.
- Promote Mr. Fuzzy.
- Improve low-performing marketing channels.
- Diversify product portfolio.
- Monitor refunds and conversion rate continuously.

15. Challenges Faced

Data modeling, DAX calculations, dashboard design, refund validation and relationship creation.

16. Skills Demonstrated

SQL, Power BI, DAX, Power Query, Data Modeling, Dashboard Design, Business Analysis.

17. Conclusion

This project demonstrates a complete SQL-to-Power BI analytics workflow delivering actionable business insights.